

Aroaro - A Tool for Distributed Immersive Mixed Reality Visualization

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ABSTRACT

In this research demo we present three immersive scenarios on three XR modalities – VR, *immersive* AR on HoloLens and 2D AR on Android. These scenarios are a network of Harry Potter characters in VR, a map-based visualization of a soldier's history with rich attributes including images and sound on HoloLens, and a visualization of car racing on Android. These visualizations have been created with Aroaro our distributed mixed reality data visualization tool.

Keywords: Augmented reality, Virtual reality, Mixed reality, Data visualization, Immersive analytics, Multi-user, Networks.

Index Terms: Human-centered computing~Visualization~Visualization application domains~Visual analytics

Human-centered computing~Human computer interaction (HCI)~Interaction paradigms~Mixed / augmented reality • Human-centered computing~Human computer interaction

1 INTRODUCTION

Aroaro, from a Māori word meaning *presence*, can be deployed on virtual reality and augmented reality devices.

Aroaro Mixed Reality (XR) app provides an immersive experience to those needing to gain knowledge by comprehending/learning from information contained in datasets, which can be explored in the exponentially amplified information space of mixed reality (Beltran et al, 2022).

Most data visualisations currently are limited by the dimensions of the screen and the clumsy affordances of keyboard and mouse.

By allowing information explorers to travel within 3D data visualizations, virtually in VR, or physically in AR, they may create many viewpoints and comments which can be explored and shared with others both synchronously and asynchronously. These may be viewed and explored either alone or as a geographically distributed group.

Mixed reality allows “physical” affordances such as ‘flying through’, ‘navigating in’, ‘surfing around’, and ‘chopping up’ the data to see information emerging from patterns created by plotting the raw data.

2 ARCHITECTURE AND ENGINEERING

Aroaro is a mini-metaverse, which - unlike those proposed by Facebook, Microsoft and others - is a relay-based peer-to-peer system. This can be setup completely under control of the owner, anyone, by getting Photon Engine and PlayFab accounts. Everyone can have their own private Aroaro spaces.

Aroaro follows five basic principles of a high-quality mixed reality environment design. It must be:

- immersive,
- easy to use,
- meaningful,
- adaptable, and
- measurable

It has been built with the best of breed industry standard components using Unity3D, PlayFab, Photon and Azure Cognitive Services. Aroaro uses Photon Engine to provide object and space synchronization. It also provides voice chat. Spatial Anchors provided by Microsoft Azure is used in Augmented Reality to synchronize objects in physical space.

Microsoft Azure Cognitive Services provides speech recognition which is important as virtual keyboards are uncomfortable to use.

Microsoft PlayFab provides a variety of backend services, which will be augmented by other services, such as image and text recognition.

3 MOTIVATION - DATA VISUALIZATION FOR MORE EFFECTIVE DECISION-MAKING

Data visualisation has made enormous progress from the time 2D images were only displayed on 2D systems (2D-on-2D). As image processing and higher quality visual display have become available, systems were able to create three-dimensional interpretations of data while displaying them on conventional screens (3D-on-2D).

With the introduction of mixed-reality technology, data visualisation now has the potential to transition to a new phase: 3D-in-3D. Our XR visualization tool processes data in a data set, turning the data points into holographic 3D objects that can be seen, manipulated, processed, and analysed while the analyst navigates in 3D space.

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The progression towards a more sensory experience of the data leads to the proposed utilisation of our XR tool as a shared data visualisation space (3D-in-3D shared). 3D-in-3D has the potential to overcome metric-driven decision-making (Tickle et al, 2016) because, when it allows for a sensory approach to data understanding and analysis (Bouyssou et al, 2013), decision makers are more exposed to the kind of thinking that drives innovation.

4 APPLICATIONS

4.1 An Organisational Perspective

Organisations want to gain insights by focusing on data analytics (Brynjolfsson et al., 2011). Nevertheless, usually an organisation can only afford that a handful of specialists can actually extract meaningful information from the data.

Data visualisation, on the other hand, can empower decision-makers with new ways of looking at, interacting with and understanding data (Olshannikova et al, 2015), (Moore, 2017). We believe our data visualisation XR prototype, running on our Aroaro environment, has the potential to contribute to improved decision making processes in organisations.

Our XR prototype relies on an innovative approach to data handling, and subsequent analysis, which can unlock the power of a sensory data approach for executives and data analysts in higher value decision-making. Aroaro is integrated with industry standard tools, specifically Power BI, which will allow an easy path for decision makers to volumetric information space.

4.2 A Child's Perspective

We have followed Alan Kay's guidance and design for children. This means that Aroaro is aimed at all who would wish to explore data. The multi-user nature of Aroaro and the ability to use it on Android (and soon iOS) devices means that groups of children could incorporate Aroaro in their learning of many subjects, from mathematics to history.

4.3 A Developer's perspective

Aroaro is being used as the basis for a mixed reality supermarket to be used by public health researchers to investigate buyer behaviour.

5 NOVELTY - DATA VISUALISATION IN A VIRTUAL REALITY CONTEXT?

Data visualization in a virtual reality context can be understood as the technology that allows us to gain richer insights into the relations amongst data points, conspicuous data sections, and hidden patterns that would not be otherwise observable with traditional 2D-based tools such as charts, graphics or tables.

With added capability, our XR tool facilitates analysts in general, and decision-makers, in particular, to move through datasets, to view data points from different angles, and to leave multi-media annotations on the data while 'traveling' the vast space occupied by the data.

The participants may explore either individually or collectively, being either co-located or remotely located. As well as projecting datasets as isolated objects of

interest, we can situate visualizations into relevant physical contexts.

6 RESEARCH DIRECTION

We report on a qualitative comparison of two visualisation platforms, Aroaro and a commercial graph visualisation tool, which were tested by volunteers (Beltran and Geng, 2021). We seek to infer how and when Aroaro is superior to the 2D flat-screen visualisation alternative.

Work will continue the development of Aroaro as a general tool for creating and evaluating visualisations in a mixed reality space. This includes work on the effectiveness of network data visualisation as a tool for immersive group decision making. In addition, Aroaro will be used in a project to create an experimental virtual reality retail space for the use of trans-disciplinary research to carry out studies on the effectiveness of product labelling and purchasing decisions from a health perspective. This will provide an MR frontend to the research focussed retail space FoodLabNZ which we are currently developing.

Underlying our research interests with mixed reality environments is the on-going issue about the effectiveness of deploying an immersive environment as an effective and superior tool to conventional approaches to data visualization, among others.

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